ARIA Visual Reference Library

Autonomous Rope Intelligence Architecture
Complete Render Documentation and Usage Guide

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March 2026

Document Overview

This document catalogs all 21 AI-generated renders produced during ARIA development (14 original concept renders + 7 patent application figures). Each render is documented with status classification, technical description, usage guidelines, and required corrections.

Purpose: Visual reference for pitch deck, YouTube series, academic paper, patent application, press kit, and CAD modeling work.

Version: 5.0 (March 2026)

Status Classification System

All renders are classified using three status labels:

Status	Meaning
[READY]	Use as-is. No corrections needed. Approved for external distribution.
[NEEDS FIX]	Correction required before external use. May be used internally. Fix instructions provided.
[REFERENCE ONLY]	Internal use only. Do not distribute externally. Superseded by later versions or contains significant errors.

Part I: Original Concept Renders (Figures 1–14)

1. Full System Cutaway

[NEEDS FIX] Centrifugal clutch label incorrect—ARIA uses ratchet/pawl mechanism. Fix label before use in academic paper or engineering documentation. Acceptable for pitch deck and YouTube with verbal correction.

Figure 1: [INSERT FIGURE 1 HERE]

ARIA full system cutaway showing internal architecture. BLDC motor with planetary gearbox, 600mm diameter rope spool, brake drum assembly, OV2640 camera lens, HX711 load cell, rope guide mechanism. STM32F411 and ESP32-S3 PCB boards visible in upper right corner. Housing rendered as transparent ghost view to show component placement.

Intended Use:

- Pitch deck slides 5 and 6 (system introduction and architecture overview)
- YouTube Episode 1 system reveal B-roll (~7:00 timestamp)
- Academic paper system overview figure (after label correction)

Fix Required: Replace "centrifugal clutch" label with "ratchet/pawl mechanism" in Canva or Illustrator before paper submission.

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2. Exploded View—Mechanism Detail

[NEEDS FIX] Duplicate labels present (cam collar labeled twice, back plate labeled twice). Components 8, 9, 11, 12 lack leader lines connecting labels to physical parts. Estimated fix time: 20 minutes in Canva/Illustrator.

Figure 2: [INSERT FIGURE 2 HERE]

ARIA mechanism exploded view showing 12 components separated along horizontal axis from housing (left) to detailed mechanism elements (right). Yellow climbing rope and weighted test bag visible.

Materials: 6061-T6 aluminum housing, 17-4PH H900 stainless steel mechanism components.

Components Correctly Labeled:

Label	Component
1	Housing (transparent ghost view—correct)
2	Back plate & wall bracket (one instance correct, one duplicate)
3	Main axle shaft (correct)
4	Rope spool with yellow climbing rope (correct, dominant visual element)
5	Ratchet wheel with asymmetric teeth (correct)
6	Flyweight assembly (correct)
7	Cam collar (one instance correct, one duplicate)
10	Pawl spring & blocker bar return spring (correct)

Required Corrections:

- Remove duplicate label 7 (cam collar)—retain upper right instance only
- Remove duplicate label 2 (back plate)—retain instance pointing to actual back plate
- Add leader lines from label 8 (blocker bar) to physical component
- Add leader lines from label 9 (pawl) to physical component
- Add leader lines from label 11 (rope guide block) to physical component
- Add leader lines from label 12 (rope guide mount bolts) to physical component
- Move labels 11 and 12 from floating position into main diagram

Fix Methodology: Export as PNG at 300 DPI, open in Canva or Adobe Illustrator, manually remove duplicate text labels, draw missing leader lines with 2pt black stroke. Do not re-prompt AI—previous iterations introduced additional errors.

Intended Use (after corrections):

- Pitch deck slide 6 (architecture component breakdown)
- Academic paper system component diagram
- Provisional patent application exploded view figure
- YouTube Episode 2 technical deep dive with component identification

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3. Ratchet/Pawl Mechanism—Iteration History

Four design iterations generated. **Use Version 3 only.** Do not extract tooth geometry or blocker bar pivot dimensions from any render—use CAD specification below instead.

Version 1—Initial Concept

[REFERENCE ONLY] Kinematics incorrect. Blocker bar shown sliding rather than pivoting. Do not use for CAD modeling.

Figure 3: [INSERT FIGURE 3 HERE]

Ratchet/pawl mechanism v1. Blocker bar shown with incorrect sliding motion (should pivot). Symmetric teeth (incorrect). Cam collar missing entirely.

Version 2—Cam Collar Added

[REFERENCE ONLY] Cam collar geometry added but duplicate labels remain. Teeth still shown as symmetric (incorrect).

Figure 4: [INSERT FIGURE 4 HERE]

Ratchet/pawl mechanism v2. Cam collar and motion arrows added. Duplicate labels present. Tooth geometry still symmetric.

Version 3—Current Reference ★

[NEEDS FIX] Best available version. Fixed pivot pin correctly labeled, duplicate labels removed. Tooth geometry still rendered symmetric—do not use render for tooth profile, refer to CAD specification below.

Figure 5: [INSERT FIGURE 5 HERE]

Ratchet/pawl mechanism v3. Fixed pivot pin labeled. Duplicate labels removed. Recommended reference for kinematic relationships. Use CAD

specification below for actual tooth geometry—render shows symmetric teeth but design requires asymmetric profile.

CAD Tooth Geometry Specification:

Parameter	Specification
Locking face angle	88° from wheel tangent (near-radial, self-energizing)
Ramp face angle	30–35° from wheel tangent
Tooth height	8–10% of wheel radius
Wheel thickness (minimum)	15 mm (SF = 5.97×)
Wheel thickness (recommended)	18 mm (SF = 7.20×)
Material	17-4PH H900 stainless, DMLS

Critical Note: The 88° locking face angle is not a preference—it is required for ANSI Z359.14 compliance. Shallow angles (8° as shown in v1) create back-drive failure mode where arrest distance approaches infinity as rope speed decreases.

Version 4—Minor Refinement

[REFERENCE ONLY] Negligible improvement over v3. Use v3 instead.

Figure 6: [INSERT FIGURE 6 HERE]

Ratchet/pawl mechanism v4. Minimal visual changes from v3.

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4. Phase 1 Prototype—Test Configuration

[READY] Usable as-is for pitch deck and YouTube content.

Figure 7: [INSERT FIGURE 7 HERE]

Phase 1 prototype in test configuration. Raw 6061-T6 aluminum housing with open face (no front cover). Yellow climbing rope loaded. Weighted sandbag visible for drop testing.

Intended Use:

- Pitch deck slide 10 (current progress/hardware development)
- YouTube Episode 1 "Where I Am Now" section
- Potential thumbnail candidate for Episode 1

Design Note: Housing renders as closed box in most views—actual design requires removable front face plate for mechanism access. Resolve disassembly sequence before finalizing housing CAD for machining.

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5. Finished Product—Wall Mounted with Climber

Three iterations progressively improving rope geometry. Version 3 closest to accurate but requires minor correction.

Version 1—No Quickdraws

[NEEDS FIX] Rope runs directly from device to climber with no intermediate quickdraws. Rope path geometry incorrect.

Figure 8: [INSERT FIGURE 8 HERE]

ARIA mounted on climbing wall with climber. Rope geometry incorrect—no quickdraws shown, rope runs directly to climber's harness.

Version 2—Zigzag Pattern

[NEEDS FIX] Rope shown threading through quickdraws but zigzags horizontally across wall face. Too many quickdraws rendered. Rope crosses itself in several locations.

Figure 9: [INSERT FIGURE 9 HERE]

ARIA with climber, improved rope path attempt. Rope zigzags laterally rather than running vertically up route. Excessive quickdraw count.

Version 3—Best Geometry ★

[NEEDS FIX] Rope geometry nearly correct—clean vertical path through 3 quickdraws. Minor rope loop artifact visible at bottom near device exit. Use inpainting to remove loop artifact—do not re-prompt entire image.

Figure 10: [INSERT FIGURE 10 HERE]

ARIA with solo climber, best rope geometry of series. Clean vertical rope through 3 quickdraws at bolts 1, 2, 3. Climber reaching for bolt 4 clip. Minor rope loop artifact at device exit requires inpainting correction. ★ Use as base for final marketing image.

Correction Method: Use Photoshop generative fill or DALL-E inpainting to remove rope loop at device exit. Select ~50px circular area around loop, prompt "smooth rope exit, no loops." Single inpainting pass should resolve.

Intended Use (after loop correction):

- Pitch deck slide 5 (primary product reveal with context)
- Marketing materials and social media
- Press kit supplementary image

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6. Finished Product—Empty Wall

[READY] Best single image in entire collection. No corrections needed. Use immediately.

Figure 11: [INSERT FIGURE 11 HERE]

ARIA mounted on lead climbing wall, no climber present. Yellow rope threaded through three quickdraws hanging slack at bolts 1, 2, 3. Blue LED status strip active at device base. Device mounted flush to wall surface (note: v5 design has shifted to behind-wall mount—this render shows earlier surface-mount concept). ★ Primary pitch image for surface-mount design phase.

Intended Use:

- Pitch deck slide 2 (The Problem—illustrating solo climber gap)
- YouTube Episode 1 cold open B-roll
- YouTube thumbnail candidate
- GitHub README header image
- High-priority distribution

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7. Product Shots—Multiple Configurations

Dark Background Studio Shot

[READY] No corrections needed. Strongest clean product image for press and marketing.

Figure 12: [INSERT FIGURE 12 HERE]

ARIA product shot on black background. Blue aluminum housing, LED status strip, mounting brackets, yellow climbing rope on spool. Professional studio lighting with soft shadows.

Intended Use:

- Pitch deck title slide or hardware section header
- Press kit primary product image (ready for immediate distribution)
- GitHub README and YouTube channel art
- Highest production value image in collection

Cutaway—Mounted Configuration

[READY] Usable as-is for technical presentations.

Figure 13: [INSERT FIGURE 13 HERE]

ARIA cutaway view mounted on climbing wall. Internal components visible through transparent housing section. First bolt and quickdraw correctly positioned.

Intended Use:

- Pitch deck slide 6 (architecture showing internals in context)
- YouTube Episode 2 technical deep dive B-roll
- Academic paper supplementary figure

Four Variant Comparison Grid

[REFERENCE ONLY] Internal comparison only. Rope geometry incorrect in all four panels.

Figure 14: [INSERT FIGURE 14 HERE]

Four-panel ARIA render comparison grid. Top left: physically accurate render. Top right: cutaway view. Bottom left: labeled component view. Bottom right: industrial design concept. All panels show incorrect rope geometry. For internal design iteration reference only.

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8. Future Render Requirements

High-Priority Corrections

- Figure 10 (Wall+climber v3)—remove rope loop at device exit via inpainting
- Figure 2 (Exploded view)—fix duplicate labels and add missing leader lines in Canva/Illustrator
- Ratchet/pawl v5—generate new render with correct asymmetric teeth (88° locking face, 30° ramp face) and unambiguous blocker bar pivot geometry

New Renders—High Priority

- Three-layer architecture diagram—concentric rings showing Mechanical (inner) / STM32 (middle) / ESP32 (outer) with failure isolation arrows
- State machine flow diagram—9 states with labeled voice command transitions and fault paths
- Night/dark gym environment—blue LED as only light source, device on standby mode
- Two-device wide shot—two ARIA units on parallel climbing routes, no climbers, showing typical gym installation

New Renders—Medium Priority

- ProGrade competitor unit—reference photo from climbingbusinessjournal.com
- TruBlue auto-belay—reference photo from trublueclimbing.com
- Lead Solo prototype—reference render from d5mag.com FIT Design Award coverage

New Renders—Lower Priority

- Phase 2 prototype—motor and electronics added to Phase 1 open housing
- Camera POV render—what OV2640 lens sees when climber reaches for bolt (clip gesture detection view)

Part II: Patent Application Figures (Figures 15–21)

Seven technical figures produced March 2026 for ARIA provisional patent application. Six hand-drawn napkin sketches + one photorealistic split-reveal render.

9. System Overview—Patent Figure 1

[READY] No corrections needed.

Figure 15: [INSERT FIGURE 15 HERE]

Hand-drawn napkin sketch on cream paper with coffee stain artifact. Side cross-section view showing rectangular housing (700×680×344 mm, 6061-T6 aluminum) mounted behind climbing wall panel. Wall bracket visible extending to structural beam. Climbing wall face shows: 60mm anodized aluminum rope port (flush mount) and LED status strip. Yellow rope threading through quickdraws QD1, QD2, QD3, QD4 to stick figure climber at bolt 5. Internal components labeled: rope spool (6061-T6, 600mm diameter), ratchet ring (200mm diameter, 30 teeth), T-Motor GB54-2 BLDC with 30:1 planetary gearbox.

Intended Use:

- Patent application Figure 1 (system overview)
- Pitch deck slides 5 and 6 (system introduction and architecture)
- Academic paper system overview figure (if informal sketch style acceptable)
- YouTube Episode 1 system reveal transition graphic

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10. Mechanical Cross-Section—Patent Figure 2

[READY] Best mechanical detail figure in entire collection.

Figure 16: [INSERT FIGURE 16 HERE]

Hand-drawn napkin sketch with detailed exploded shaft view and cross-hatching. Main shaft assembly showing (left to right): ratchet ring (200mm diameter, 30 teeth, 17-4PH H900 DMLS, 18mm thick, 88° locking face / 32° ramp face with asymmetric tooth profile), dual pawls (A2 tool steel 58 HRC), blocker bar (17-4PH H900), rope spool (6061-T6 aluminum, hub diameter 120mm, flange diameter 240mm), INA HFL2026 one-way sprag bearing (20mm ID × 47mm OD—annotated "motor CANNOT backdrive catch"), T-Motor GB54-2 BLDC with 30:1 gearbox. Flyweight mechanism shown in circular detail inset: cylindrical slug 20mm diameter × 58mm length, mass 141g, pivot radius r=45mm, compression spring preload 2.5N. Main shaft: 4140 heat-treated steel, 25mm diameter. Extensive cross-hatching for metal surfaces.

Intended Use:

- Patent application Figure 2 (mechanical cross-section and detailed mechanism view)
- Academic paper mechanical subsystem detailed figure
- YouTube Episode 2 technical deep dive with component identification overlay
- Highest technical detail render—primary engineering reference

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11. Electronics Architecture—Patent Figure 3

[NEEDS FIX] Confidence threshold values show >2.85 and >2.75 (incorrect—should be ≥0.85 and ≥0.75). STM32 block missing "9-STATE MACHINE" and "PID TENSION 40N" functional boxes.

Figure 17: [INSERT FIGURE 17 HERE]

Hand-drawn napkin sketch showing two-layer electronics architecture. Top: STM32F411 Black Pill safety layer ($8 cost labeled). Bottom: XIAO ESP32-S3 Sense intelligence layer ($20 cost labeled). Double-headed arrow between layers labeled "UART 115200 baud." Left side inputs to

STM32: HX711 load cell module (50kg capacity), AS5048A magnetic encoder (14-bit SPI), E-STOP button (hardware interrupt). Left side inputs to ESP32: OV2640 camera (built-in to ESP32-S3 Sense), PDM microphone (built-in). Right side output: VESC Mini 6.7 motor controller connected to T-Motor GB54-2 (hand-drawn motor diagram). Bottom annotation: "if ESP32 crashes → STM32 + motor keep running—mechanical catch still works." Confidence thresholds shown incorrectly as ">2.85 conf" and ">2.75 conf" (should be "≥0.85" and "≥0.75").

Required Corrections (Canva/Illustrator):

- Paint over ">2.85 conf" text → replace with "≥0.85 conf"
- Paint over ">2.75 conf" text → replace with "≥0.75 conf"
- Add missing functional box to STM32 block: "9-STATE MACHINE"
- Add missing functional box to STM32 block: "PID TENSION 40N"
- Remove duplicate "FAULT WATCHDOG 1020ms" box if present

Intended Use (after corrections):

- Patent application Figure 3 (electronics architecture)
- Academic paper two-layer safety architecture diagram
- Pitch deck slide 6 electronics detail overlay
- YouTube Episode 2 software architecture explanation

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12. State Machine—Patent Figure 4

[NEEDS FIX] Duplicate "CLIMBING PAUSED" box visible on left side (remove lower instance). Garbled box in lower center should be labeled "WATCH_ME—motor: TENSION 60N."

Figure 18: [INSERT FIGURE 18 HERE]

Hand-drawn napkin sketch showing 9-state finite state machine diagram. States: IDLE, CLIMBING (center position with double border emphasis), CLIPPING, TAKE, REST, LOWER, WATCH_ME, UP, CLIMBING PAUSED, ESTOP. Red dashed arrows from every state to ESTOP labeled "e-stop button or fault." Bottom annotation: "TAKE is the only safety-critical command—requires 2-factor confirmation." Duplicate CLIMBING PAUSED box visible on left (remove lower instance). Garbled box in lower center should read "WATCH_ME—motor: TENSION 60N" not "TAKE."

Required Corrections (Canva/Illustrator):

- Delete lower duplicate "CLIMBING PAUSED" box—retain upper-left instance only
- Relabel garbled lower box from "TAKE" to "WATCH_ME—motor: TENSION 60N"

Intended Use (after corrections):

- Patent application Figure 4 (state machine diagram)
- Academic paper state machine and control logic figure
- Firmware documentation reference diagram

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13. Two-Factor TAKE Timing—Patent Figure 5

[READY] No corrections needed.

Figure 19: [INSERT FIGURE 19 HERE]

Hand-drawn timing diagram with 4 horizontal rows showing 2-factor voice command verification sequence. Row 1 (VOICE): spike at t=0.3s showing confidence score 0.91 (above ≥0.85 threshold line). Row 2 (CONFIRMATION WINDOW): hatched rectangle from t=0.3s to t=0.8s (500ms window duration). Row 3 (LOAD CELL force in Newtons): 40N baseline tension with sharp spike to 250N at t=0.65s, exceeding 200N threshold dashed line. Row 4 (STATE): showing CLIMBING → CLIMBING (awaiting confirm) → TAKE transition. Failure case annotated at right: "voice command at t=0.3s, load cell never exceeds 200N, confirmation window expires at t=0.8s, state remains CLIMBING—command rejected."

Intended Use:

- Patent application Figure 5 (two-factor timing diagram)
- Academic paper two-factor verification protocol figure
- Pitch deck slide on safety architecture details
- YouTube Episode 3 (software safety mechanisms)

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14. Fault Hierarchy—Patent Figure 6

[NEEDS FIX] POWER_LOW fault (<11V → 30% current limit) missing from Tier 3. Add as 5th item after ESP32_CRASH in Canva.

Figure 20: [INSERT FIGURE 20 HERE]

Hand-drawn napkin sketch showing 4-tier fault hierarchy pyramid. TIER 1 (red shading) ESTOP—immediate motor disable and catch arming: ENC_DROPOUT, MOTOR_FAULT, MOTOR_BACKDRIVE, STM32_WATCHDOG, POWER_CRITICAL. TIER 2 (amber shading) HOLD—maintain current position: LC_DROPOUT, LC_FROZEN, ENC_JUMP. TIER 3 (tan shading) DEGRADE—reduced functionality: LC_OVERRANGE, MOTOR_THERMAL, UART_TIMEOUT, ESP32_CRASH (note: POWER_LOW missing from this tier—add in Canva). TIER 4 (gray shading) IGNORE—log but continue: UART_CORRUPT. Bottom annotation with emphasis box: "MECHANICAL CATCH—completely independent of ALL electronics—armed under every fault condition." Margin note in different handwriting: "UART_TIMEOUT is key design decision—ESP32 can crash completely and STM32 + motor keep running—climber never knows intelligence layer failed."

Required Corrections (Canva/Illustrator):

- Add to Tier 3 after "ESP32_CRASH": "POWER_LOW (<11V → 30% current)"

Intended Use (after corrections):

- Patent application Figure 6 (fault hierarchy and response classification)
- Academic paper fault response table
- Dashboard software "Fault Behavior" tab reference diagram
- System safety documentation

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15. Split-Reveal: Behind-Wall Mount ★

[READY] Strongest render produced in March 2026 design session. No corrections needed. Use immediately as primary pitch image.

Figure 21: [INSERT FIGURE 21 HERE]

Split-panel photorealistic render with diagonal division. LEFT PANEL (climbing wall face view): textured climbing wall surface with yellow dynamic rope running vertically through quickdraws at bolts 1, 2, 3. iPad HMI mounted on low-profile bracket showing ARIA Climb app interface—status: ACTIVE, tension: 42N, state: CLIMBING with three command buttons (TAKE / SLACK / REST). ARIA wordmark laser-etched on wall surface below iPad. Blue LED status strip (80mm length) at base glowing active. 60mm diameter anodized aluminum rope port (flush mount with smooth radius entry). RIGHT PANEL (cutaway behind-wall view): wall panel rendered transparent to reveal blue ARIA housing (700×680×344mm) mounted on structural timber beam via bolt bracket. Yellow rope visible on spool inside housing. PCB boards (STM32 + ESP32) visible in upper corner. Ratchet mechanism components visible. Diagonal split line between panels creates dramatic reveal effect. Single image communicates complete design philosophy: device hidden behind wall, only rope port, LED strip, and HMI visible to climber.

Intended Use:

- Pitch deck primary reveal slide (highest priority—use as main"what is ARIA" visual)
- YouTube Episode 1 thumbnail candidate (strong visual impact)
- Press kit hero image (ready for immediate distribution)
- GitHub README header image
- Academic paper system overview figure (if photorealistic style acceptable)
- Social media launch graphics

Design Significance: This render most effectively communicates the March 2026 behind-wall mounting decision. Previous renders (Figures 1–14) show surface-mount design. Figure 21 is first render accurately depicting behind-wall configuration with flush rope port and hidden housing.

Part III: Usage Guidelines and Quick Reference

Status Summary Table

Figure	Status	Primary Use
1	NEEDS FIX	Pitch deck (fix label first)
2	NEEDS FIX	Patent/paper (fix labels first)
3, 4, 6	REFERENCE ONLY	Do not use externally
5	NEEDS FIX	CAD reference (use spec not render)
7	READY	Pitch deck slide 10, YouTube Ep 1
8, 9	NEEDS FIX	Do not use (rope geometry wrong)
10	NEEDS FIX	Marketing (inpaint loop first)
11	READY	Pitch deck slide 2, YouTube, README
12	READY	Press kit, title slide
13	READY	Pitch deck slide 6, YouTube Ep 2
14	REFERENCE ONLY	Internal iteration only
15	READY	Patent Fig 1, pitch deck
16	READY	Patent Fig 2, paper (TOP PRIORITY)
17	NEEDS FIX	Patent Fig 3 (fix confidence values)
18	NEEDS FIX	Patent Fig 4 (remove duplicate)
19	READY	Patent Fig 5, paper
20	NEEDS FIX	Patent Fig 6 (add POWER_LOW)
21	READY	Pitch deck hero image (TOP PRIORITY)

Use Immediately (No Corrections Required)

- **Figure 21**—Split-reveal behind-wall mount: pitch deck hero slide, YouTube thumbnail, README header, press kit [HIGHEST PRIORITY]
- **Figure 16**—Mechanical cross-section napkin: patent Fig 2, academic paper detailed mechanism figure [HIGHEST PRIORITY]

- **Figure 11**—Empty wall with rope: pitch deck slide 2, YouTube B-roll, GitHub README
- **Figure 12**—Dark background product shot: press kit, pitch deck title slide
- **Figure 13**—Cutaway on wall: pitch deck slide 6 architecture detail
- **Figure 7**—Phase 1 prototype: pitch deck slide 10, YouTube Episode 1
- **Figure 15**—System overview napkin: patent Fig 1, pitch deck slides 5-6
- **Figure 19**—Two-factor timing diagram: patent Fig 5, academic paper safety protocol

Fix Then Use (5–20 Minutes Per Figure)

Figure	Fix Required	Tool	Time
17	Confidence values + STM32 boxes	Canva	5 min
18	Remove duplicate CLIMBING PAUSED	Canva	3 min
20	Add POWER_LOW to Tier 3	Canva	2 min
10	Inpaint rope loop at device exit	Photoshop	5 min
2	Fix duplicate labels + leader lines	Illustrator	20 min

Do Not Use Externally

- Figures 3, 4, 6 (Ratchet/pawl v1/v2/v4)—kinematics incorrect, superseded by v3
- Figures 8, 9 (Wall+climber v1/v2)—rope geometry fundamentally wrong
- Figure 14 (Four-panel comparison)—rope geometry wrong in all panels

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Part IV: Production Notes

Render Generation Timeline

- **December 2025**—Figures 1–14 (original concept renders), iterative refinement via DALL-E 3
- **March 2026**—Figures 15–21 (patent application figures), napkin sketch style + split-reveal photorealistic

File Organization

All source renders stored at 1024×1024 resolution (square) or 1024×1536 resolution (portrait). Export at 300 DPI for print materials. Original PNG files retained with alpha channel for compositing.

File naming convention:
`ARIA_render_[number]_[description]_[version].png`

Example: `ARIA_render_21_split_reveal_v1.png`

Version Control

This document is version 5.0. Previous versions:

- v1 (December 2025)—14 original renders only
- v2 (January 2026)—Added ratchet/pawl iterations
- v3 (February 2026)—Updated status classifications
- v4 (March 2026)—Added patent figures 15–20
- v5 (March 2026)—Added figure 21, complete usage guidelines

Color and Branding

ARIA brand colors:

- Housing: Blue (#2E5090) aluminum powder coat
- LED status: Blue (#00A8E8) active, amber (#FFB703) caution, red (#DC143C) fault
- Rope: Yellow (#FFD60A) standard dynamic climbing rope
- Accent: White (#FFFFFF) for text and HMI elements

Conclusion

This reference library documents 21 renders across 15 months of iterative design visualization. Nine renders are ready for immediate external use (Figures 7, 11, 12, 13, 15, 16, 19, 21, and corrected 20). Five require minor corrections (5–20 minutes each). Seven are superseded and should not be used externally.

Top priority renders for March 2026:

1. Figure 21 (split-reveal)—primary pitch deck hero image
2. Figure 16 (mechanical cross-section)—patent and academic paper
3. Figure 11 (empty wall)—marketing and README

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Last updated: March 7, 2026